

Yellow Jacket Swarm Counter Unmanned Aerial System



DIU Counter UAS Sensing for Homeland and Mobile Defense (PROJ00656)

Interstellar Foundry retains commercial IP for the core sensor-fusion algorithm, while granting the DoD Government Purpose Rights to specific interface APIs and ASCII outputs

The Team



- **Company History & Financing:** An early-stage defense technology team founded in January 2026 incubated within the **Georgia Tech Create-X** startup program. Currently operating with pre-seed academic grant funding and institutional prototyping support.
- **Size:** 6 Full-Time Project Members.
- **CAGE Code / UEI:** Pending
- **Key Personnel:**
 - Parth Patel (Mechanical Engineer)
 - Calkin Garg (Computer Science)
 - Zheng-Yin Lee (Electrical/Computer Engineer)
 - Ethan Yee (Mechanical Engineer)
 - Carlos Cordova (Electrical/Computer Engineer)
 - Rashod Abdurasulov (Computer Science)

The Threat · Why Now

"Today, autonomy thrives; tomorrow, intelligence prevails."



Low-RCS, Low-Altitude

Commercial drones fly under legacy radar thresholds and Small cross-sections evade S-band sensors designed for aircraft.



Sustainment Cost

Current COTS counter-drone solutions cost \$50K-\$500K per node. Cannot scale to protect dispersed DoD infrastructure.



Fixed-Wing Gap

Fixed-wing interceptors lack agility to track quadcopters making abrupt evasive maneuvers at 25+ mph in close engagements.



OUR SOLUTION

Sub-\$1,000 modular ground station

- ✓ NVIDIA Jetson Nano edge computing
- ✓ FMCW radar + CV sensor fusion
- ✓ Kalman filter → eliminates false positives
- ✓ 3× autonomous quadcopter interceptors
- ✓ Waterjet CF frames → < \$280/unit
- ✓ < 15 min deployment on ISV/JLTV

Georgia Tech Create-X · DIU PROJ00656

System Architecture · 4-stage Pipeline

01



Primary Detection

RADAR

24 GHz FMCW
K-band Doppler
Range + velocity
≤20 dBm emission

02



Secondary Tracking

OPTICAL

OAK-D Pro stereo
Hikvision 8MP
Angular precision
Object classification

03



Edge Compute

JETSON

NVIDIA Jetson Nano
60+ TOPS, 7-25W
YOLOv8 inference
No cloud dependency

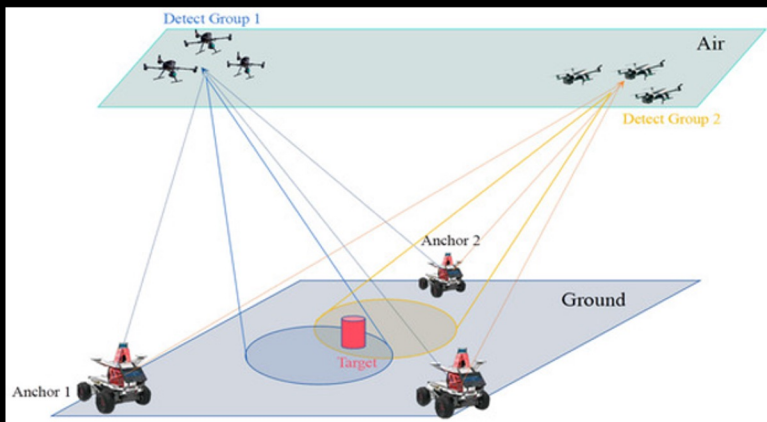
04



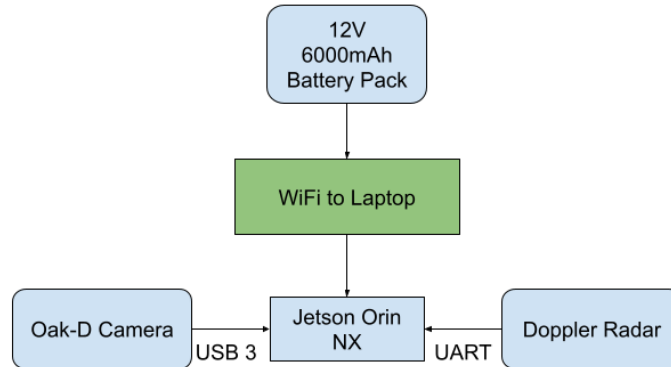
Sensor Fusion

HANDSHAKE

Kalman filter
Fuses radar + optics
Elim. false positives
Locks Group 1/2 sUAS



Field-tested Casing



Sensor Fusion Physics · Kalman Filter

The math that locks Group 1/2 targets while eliminating birds and weather clutter

RADAR DETECTION

Doppler frequency shift:

$$f_D = 2v_r \cdot f_c / c$$

Radar range equation:

$$R = [(P_t \cdot G_t \cdot G_r \cdot \lambda^2 \cdot \sigma) / ((4\pi)^3 \cdot S_{min})]^{(1/4)}$$

FMCW range resolution:

$$\Delta R = c / (2 \cdot B) \quad \text{where } B = \text{bandwidth}$$

Velocity resolution:

$$\Delta v = \lambda / (2 \cdot T_{CPI})$$

f_D : Doppler frequency shift, v_r : Relative velocity, f_c : Carrier frequency,
 c : Speed of light, R : Target range, P_t : Transmit power, G_t : Transmit antenna gain,
 G_r : Receive antenna gain, λ : Wavelength, σ : Radar cross section,
 S_{min} : Minimum detectable signal, ΔR : Range resolution, B : Bandwidth,
 Δv : Velocity resolution, T_{CPI} : Coherent processing interval

360°

Azimuth Coverage

10 Hz

Update Rate

KALMAN FILTER FUSION

State prediction:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F} \cdot \hat{\mathbf{x}}_{k-1} + \mathbf{B} \cdot \mathbf{u}_k$$

Covariance prediction:

$$\mathbf{P}_{k|k-1} = \mathbf{F} \cdot \mathbf{P}_{k-1} \cdot \mathbf{F}^T + \mathbf{Q}$$

Kalman gain:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \cdot \mathbf{H}^T \cdot (\mathbf{H} \cdot \mathbf{P}_{k|k-1} \cdot \mathbf{H}^T + \mathbf{R})^{-1}$$

State update:

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \cdot (\mathbf{z}_k - \mathbf{H} \cdot \hat{\mathbf{x}}_{k|k-1})$$

$\hat{\mathbf{x}}_{k|k-1}$: Predicted state estimate, $\mathbf{F}$: State transition model, $\mathbf{z}_k$: Measurement vector
 $\hat{\mathbf{x}}_{k-1}$: Previous state estimate, $\mathbf{B}$: Control input model, $\mathbf{u}_k$: Control vector,
 $\mathbf{P}_{k|k-1}$: Predicted covariance estimate, $\mathbf{Q}$: Process noise covariance, $\mathbf{K}_k$: Kalman gain,
 $\mathbf{H}$: Observation model, $\mathbf{R}$: Measurement noise covariance, $\hat{\mathbf{x}}_k$: Updated state estimate

<5 mrad

Az/El Error

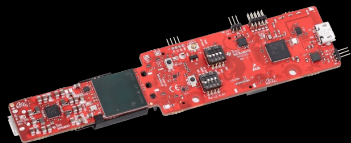
50 mph Vehicle Track Speed

Hardware Subsystems

Radar (Tiered Scaling) —>



TI IWR6843AOPEVM (~\$220)
for high-speed DSP and
>50m tracking.



Optical Suite —>



Highly compatible with
OAK-D Pro (RGB/stereo
depth) or



Edge Processing —>



NVIDIA Jetson Nano delivers
localized, cloud-independent AI
(60+ TOPS) on a highly efficient
7-25W power draw.



Tactical Footprint —>



Sub-10 lb modular design
supports under 15-minute
deployment on ISV, JLTV,
FMTV, and HEMTT platforms.



Software Subsystems

GNSS-Denied Navigation —>



Converts 2D pixels to 3D map
coordinates using OpenCV and
SLAM to localize threats
without GPS.

Spectral Stealth —>



Air-gapped serial communication
and passive optics ensure maximum
EMI robustness and low
physical/spectral signatures.

Cybersecurity



Architecture strictly complies
with **DOD Instruction 8510.01**
(RMF) to streamline the
Authorization to Operate (ATO).

LOE 1 Homeland

ASCII output · FAAD C2 · 5-30kW gen compatible

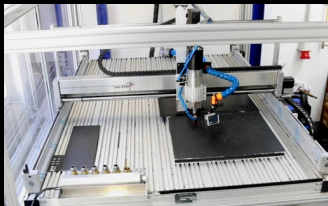
LOE 2 Tactical/Mobile

GNSS-denied · ISV/JLTV mount · passive posture

UAV Hardware · Manufacturing Approach

HARDWARE SPECS

Frame	450mm quad · waterjet-cut carbon fiber · 3D-printed mounts
Motors	2204 2300KV brushless x 4 per airframe
Props	5" balanced props · optimized for 80+ km/h intercept
Flight Ctrl	Matek F405 running ArduCopter firmware
Comms	ESP32 WiFi MAVLink telemetry to ground station
Cost	~\$280/unit · ~\$1,617 total for 3 swarm C-UAS
Weight	< 10 lb deployable ground station
Deploy	< 15 min setup



DESIGN FOR MANUFACTURABILITY

Waterjet-Cut CF Frames

2D DXF files → waterjet table. Minimal downtime. Consistent tolerance ± 0.005 "

3D-Printed Mounts & Arms

PETG/ABS mounts for motors, ESCs, sensors. Rapidly iterate design.

COTS Avionics Stack

Matek F405 + ESP32: proven, in-stock, programmable. No defense foundry lead times. ATO-streamlined.

Cost Scalability

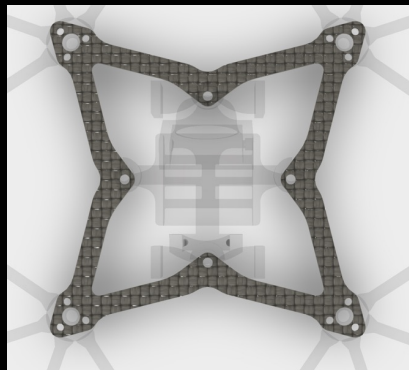
Tier 1: ~\$280/unit MVP.
Tier 2 (6-25): Low-Rate Production.
Tier 3 (26-100+): mass manufacturing.

Interceptor Design · CAD Modeling

Airframe: 450mm quadcopter configuration. Waterjet-cut carbon fiber main plate (3K twill weave, 3mm thickness). CAD modeled in Fusion 360. 3D-printed PETG motor mounts and camera bracket assembly.

Propulsion: 2204 2300KV brushless motors × 4, 5" balanced propellers, optimized for 80+ km/h intercept.

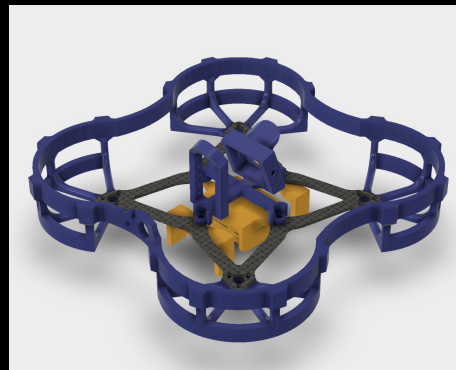
Flight Control: Matek F405 running ArduCopter 4.3. ESP32 MAVLink telemetry bridge. Coordinated engagement vectors uplinked from ground station via Python API.



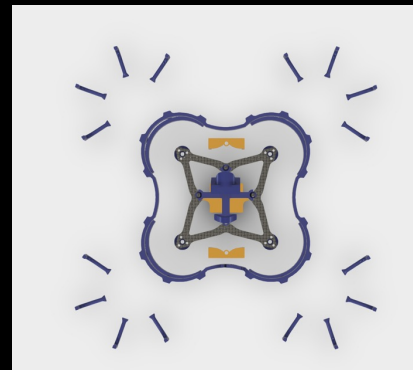
CF Frame (top view)
Waterjet DXF → waterjet table



Payload + Camera Mount
3D-printed PETG



Full Assembly: CF frame (gray) +
guards (blue) + mounts (gold)



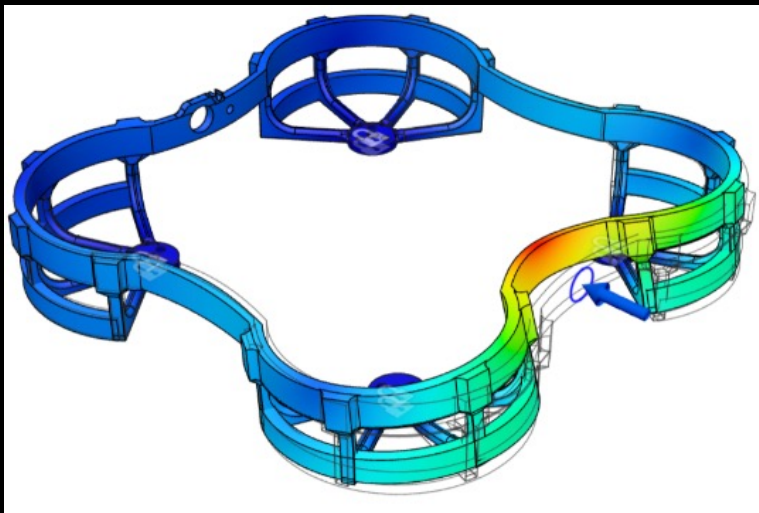
Full Assembly Exploded View

Material Selection · Structural FEA

Test Campaign: 10 prop guard specimens across 5 filament types (PETG-CF, CF-PLA, CF-Nylon, LW-PLA, Polycarbonate) loaded to failure under axial compression. Peak load plotted over time to characterize both maximum strength and failure mode (brittle fracture vs. progressive deformation).

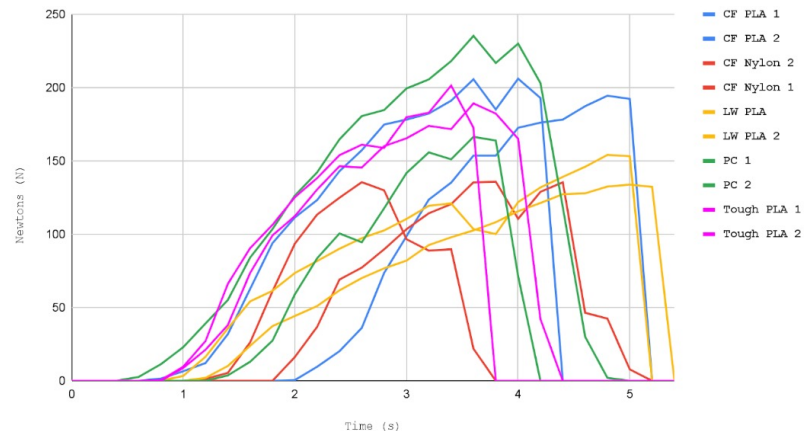
Key Finding: CF-PLA achieved peak loads exceeding 220N but fractured catastrophically. Polycarbonate showed the best post-failure behavior (plastic deformation rather than shattering), selected as primary material for guards.

$$\text{Von Mises Stress: } \sigma_{vm} = \sqrt{(\sigma_{\theta}^2 + \sigma_{ax}^2 - \sigma_{\theta} \cdot \sigma_{ax})}$$



ANSYS FEA: Prop guard structural analysis
Von Mises stress distribution under axial compression load.

Material Test Results



Material test result: Force (N) vs. time (s) across 10 specimens.

Physical Build Setup

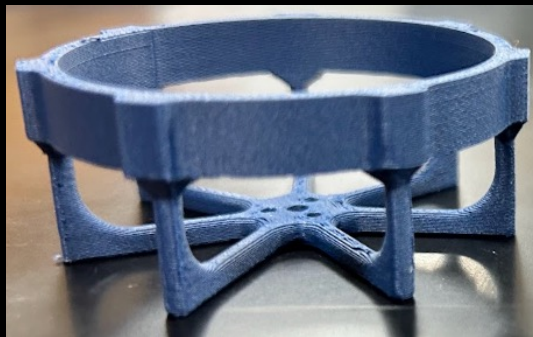
Carbon Fiber Main Plate: 2D DXF exported from Fusion 360 and cut on a waterjet table (± 0.005 " tolerance).

3D-Printed Components: Motor mounts, prop guards, and camera brackets printed in PETG and polycarbonate using Bambu Lab X1 Carbon. Iterated across 5 revs to achieve target budget and impact resilience.

Assembly Budget: ~\$280/interceptor. Total 3-unit CUAS: ~\$1,617.



Waterjet CF frame
(physical part)



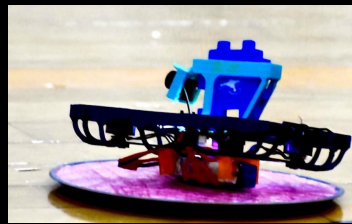
3D-printed motor guards



3D-printed prop guards



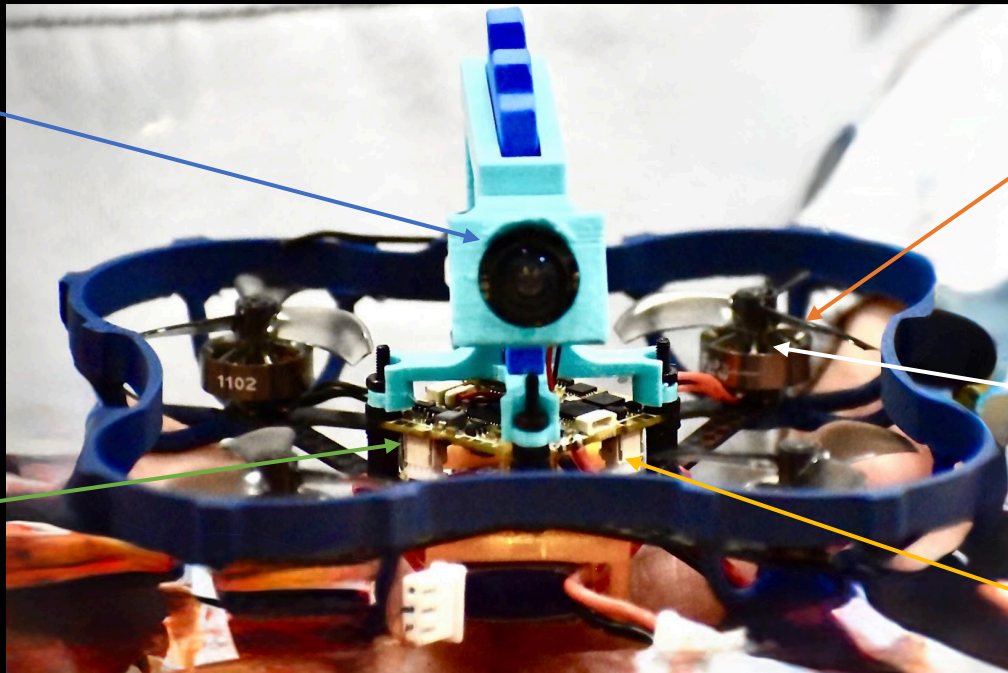
Camera/Payload mount
finished part



Bee Eye



BLV4



Flow 1102



Azi Triblade



XT-30

Engagement Logic · Swarm Software

GROUND STATION SOFTWARE

Python / OpenCV

Real-time video processing pipeline

HB100 Doppler

Velocity + range primary detection

Kalman (FilterPy)

Multi-target tracking + fusion

YOLOv8 inference

Jetson-accelerated classification

PyQt6 UI

Operator display / engagement cue

ENGAGEMENT LOGIC

Target acquired

Dual-sensor confirm → lock

Primary dispatch

Lead interceptor → head-on vector

Flankers launch

2 flankers on diverging intercept

Abort logic

Flankers abort if primary scores
kill

Re-engage

Missed intercept → re-cue loop

MVP SUCCESS CRITERIA

Dual-sensor detect

Both radar AND camera confirm

80+ km/h targets

Intercept at full speed

3 simultaneous launches

Coordinated swarm execution

Kinetic contact

Physical intercept confirmed

< 15 min deploy

From packed to combat ready

Business Case · Current Market

The gap: low-cost, high-lethal, mass-producible C-UAS at scale

TIERED ROM PRICING

TIER 1 1-5 units

\$500-\$1,000/node

Prototype / evaluation

TIER 2 6-25 units

LRIP pricing

Low-rate initial production

TIER 3 26-100+ units

Mass manufacturing

Full contract scale

MARKET CONTEXT

\$4B+

C-UAS global market by 2030

~\$500K

Avg. legacy sensor node cost

<\$1K

Our node cost is 500× cheaper

Group 1/2

Primary threat is underserved

Point of Contact: Parth Patel

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